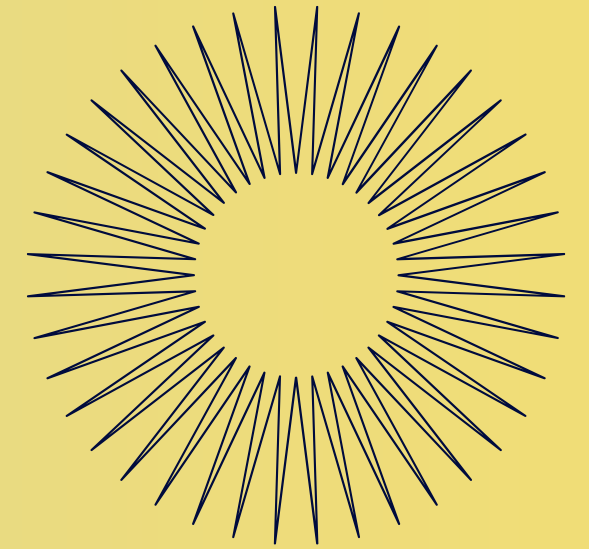


ITC HOTELS REVENUE OPTIMIZATION DASHBOARD



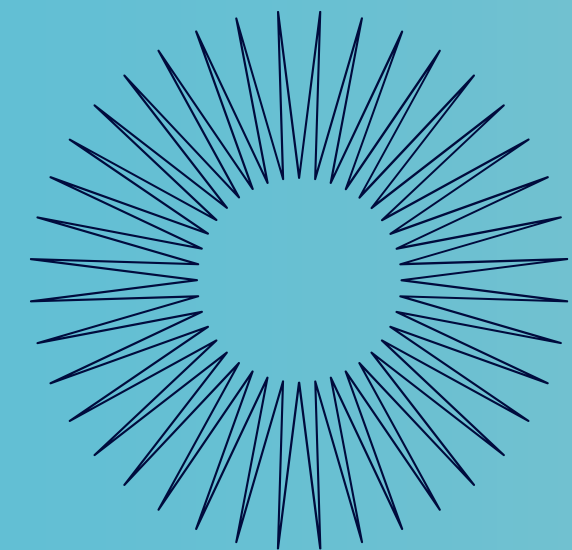
LUXURY HOTELS OF ITC:
DATA-DRIVEN INSIGHTS &
PERFORMANCE
DASHBOARD BY USING
POWER BI

presented by : Atul Soni

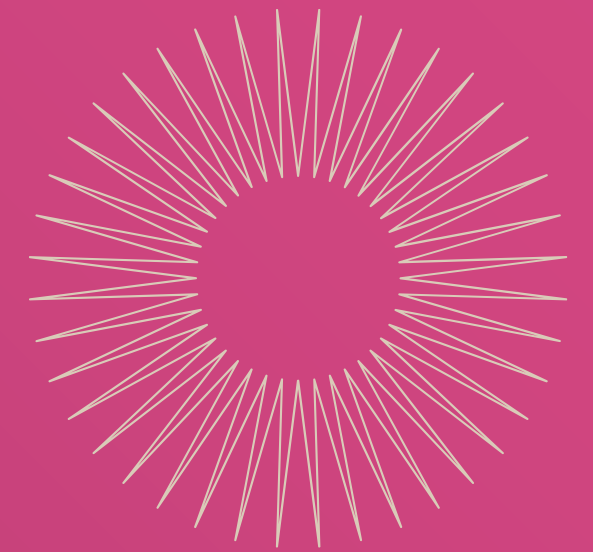


ITC HOTEL HISTORY

- ITC LIMITED ENTERED THE HOTEL BUSINESS ON 18 OCTOBER 1975 WITH THE OPENING OF A HOTEL IN CHENNAI.
- ITC HOTELS CURRENTLY OWN AND OPERATE 100 HOTELS IN 75 DESTINATIONS.
- INDIA'S THIRD-LARGEST HOTEL CHAIN.
- THE GROUP TODAY OPERATES UNDER DIFFERENT BRANDS:
- ITC LUXURY COLLECTION HOTELS
- WELCOME HOTEL SHERATON HOTELS
- FORTUNE HOTELS
- WELCOME HERITAGE HOTELS
- WELCOME ENVIRON



Agenda Overview



Content: -

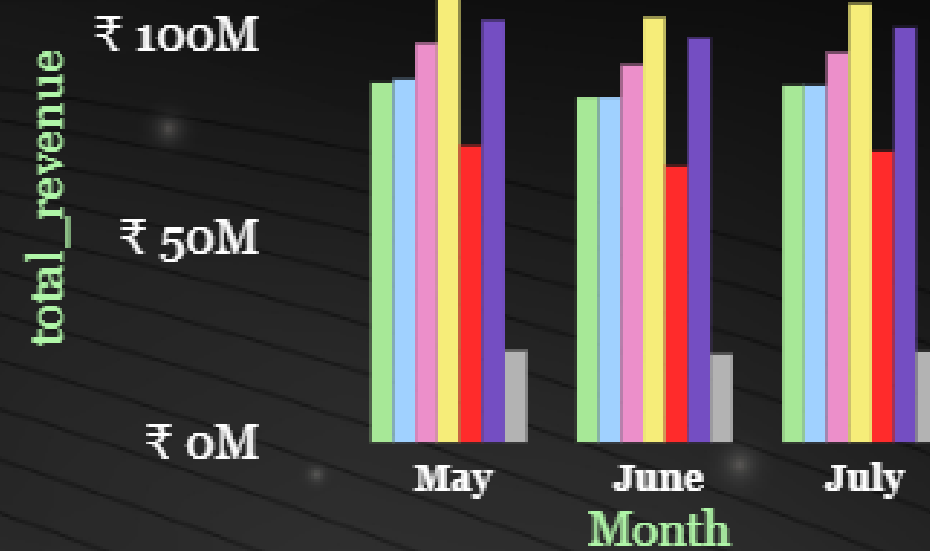
- Business Context & Objectives
Understanding the problem and project goals
- Data Overview & Methodology
Dataset structure and approach used
- Financial Performance Analysis
Revenue trends, ADR, and RevPAR insights
- Occupancy & Capacity Analysis
Utilization patterns and demand trends
- Room Category & Booking Insights
Customer behavior and room performance
- Cancellation & Revenue Impact
Cancellation trends and lost revenue
- Key Insights & Strategic Recommendations
Actionable business decisions



Financial Overview & Revenue Performance

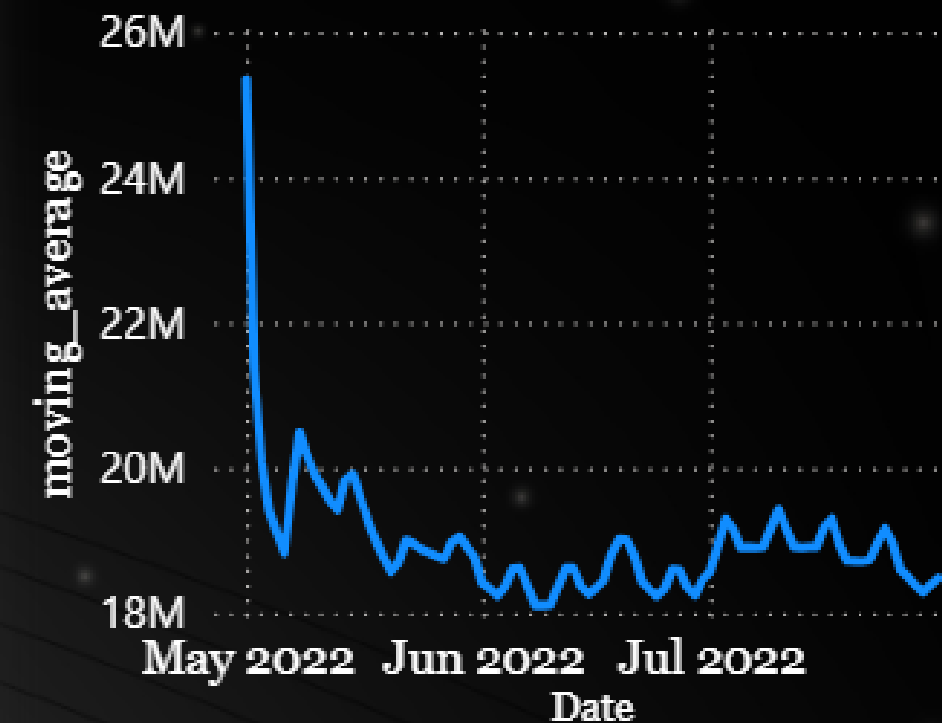
Total_revenue by Month and property_name

property... ● ITC Bay ● ITC Blu ● ITC City ▶



Trends of paramters

moving_average and total_revenue by Date



Charts

Parameter

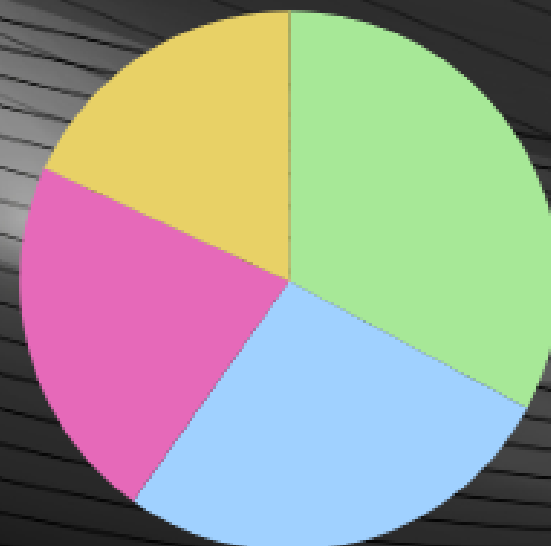
moving_average

Running total

Date	running_total_revenue
01 May 2022	₹2,53,57,111
02 May 2022	₹4,27,65,917
03 May 2022	₹6,03,00,105
04 May 2022	₹7,76,32,159
05 May 2022	₹9,55,33,195
06 May 2022	₹11,29,60,081
Total	₹1,70,87,71,229

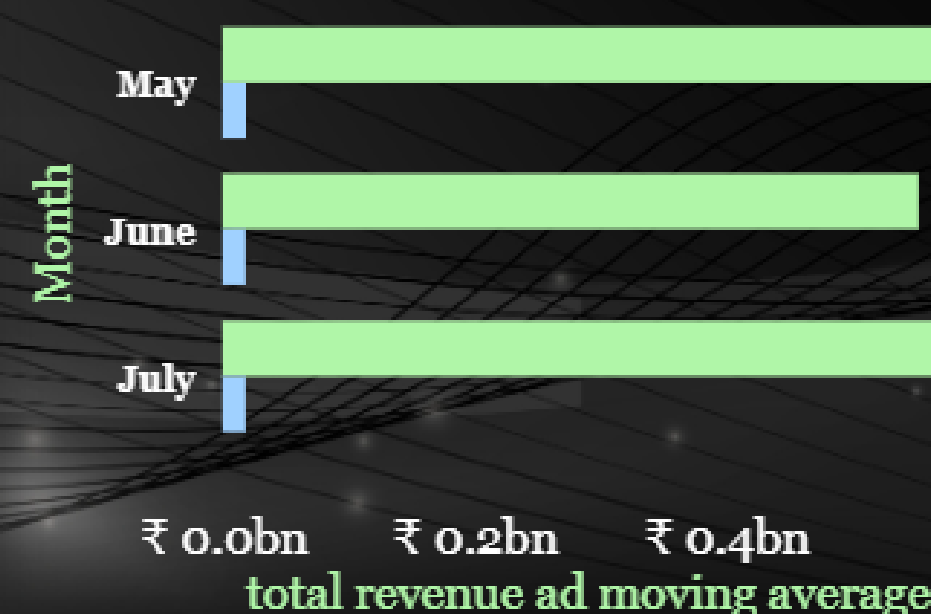
Total_revenue by room_class

room_cl... ● Elite ● Premium ● Presidential ▶



Moving average by month

● total_revenue ● moving_average



Hotel names

All

₹ 1.71bn

total_revenue



50.44%

mom_growth

14.92K

ADR

7.35K

RevPAR

₹2bn

running_total_revenue

Room category

All

Date slicer

All

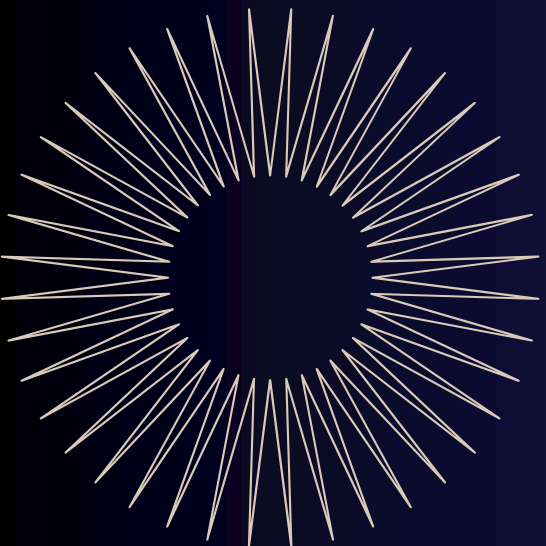


Revenue Analysis



Quantitative Insights – Financial Overview :-

- TOTAL REVENUE: ₹1.71B GENERATED, WITH CUMULATIVE REVENUE CROSSING ₹2B, INDICATING STRONG OVERALL BUSINESS SCALE.
- GROWTH TREND: REVENUE INCREASED FROM ~₹90–100M IN MAY TO ₹110–120M IN JULY PER PROPERTY, REFLECTING A ~15–25% MONTHLY INCREASE.
- MOM GROWTH: 50.44% GROWTH HIGHLIGHTS A SHARP RISE IN PERFORMANCE, LIKELY DRIVEN BY HIGHER DEMAND OR PRICING IMPROVEMENTS.
 - REVENUE CONTRIBUTION: PREMIUM & ELITE ROOMS CONTRIBUTE ~60–70% OF TOTAL REVENUE
 - BASE CATEGORIES CONTRIBUTE <40%, SHOWING IMBALANCE IN REVENUE DISTRIBUTION
 - PRICING VS PERFORMANCE: ADR: ₹14.92K (STRONG PRICING LEVEL)
 - REVPAR: ₹7.35K → IMPLIES ~49% EFFECTIVE OCCUPANCY ($\text{REVPAR} \div \text{ADR}$)
- TREND STABILITY: MOVING AVERAGE FLUCTUATES WITHIN ₹18M–₹25M RANGE, INDICATING CONTROLLED VARIATION WITHOUT MAJOR VOLATILITY
- CUMULATIVE GROWTH: RUNNING TOTAL SHOWS A STEADY RISE, ADDING ₹70M–₹90M DAILY, REFLECTING CONSISTENT BOOKING INFLOW.



Occupancy & Capacity Analysis

43.50%

occupancy_rate

13.14%

occupancy_wow_gr...

233K

total_room_available

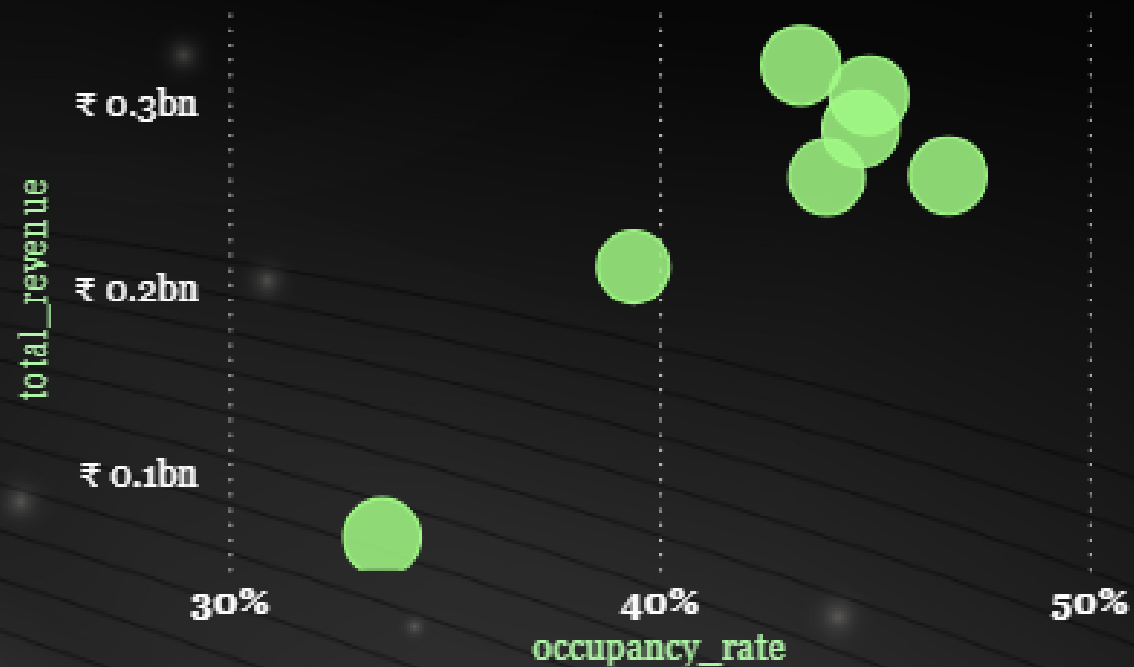
-0.12%

mom_occupancy_gr...

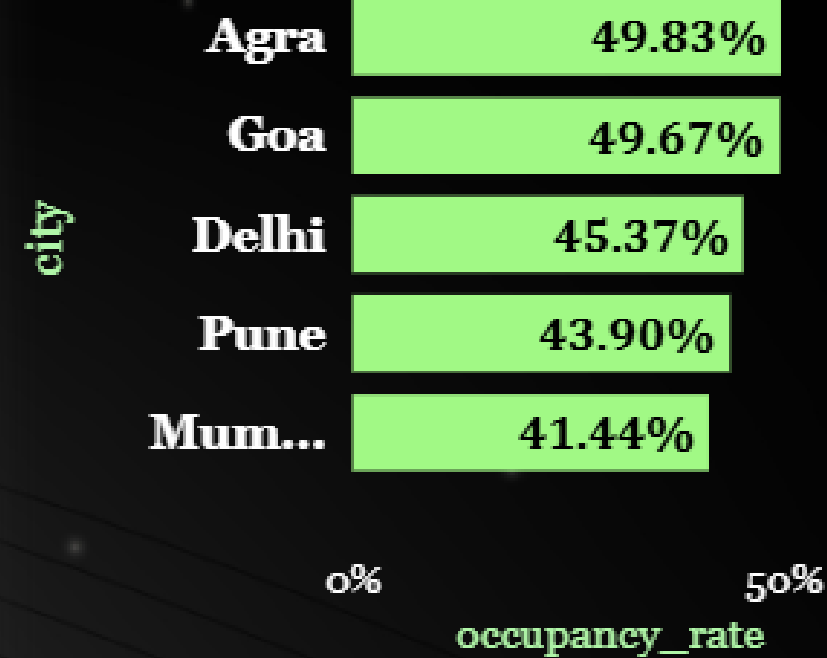
7.35K

RevPAR

Occupancy correlate with revPAR and total revenue



occupancy_rate by city



occupancy_rate

Parameter

room_class

city

property_name

category

Date slicer

01-05-2022

31-07-2022

Hotel names

All

Room category

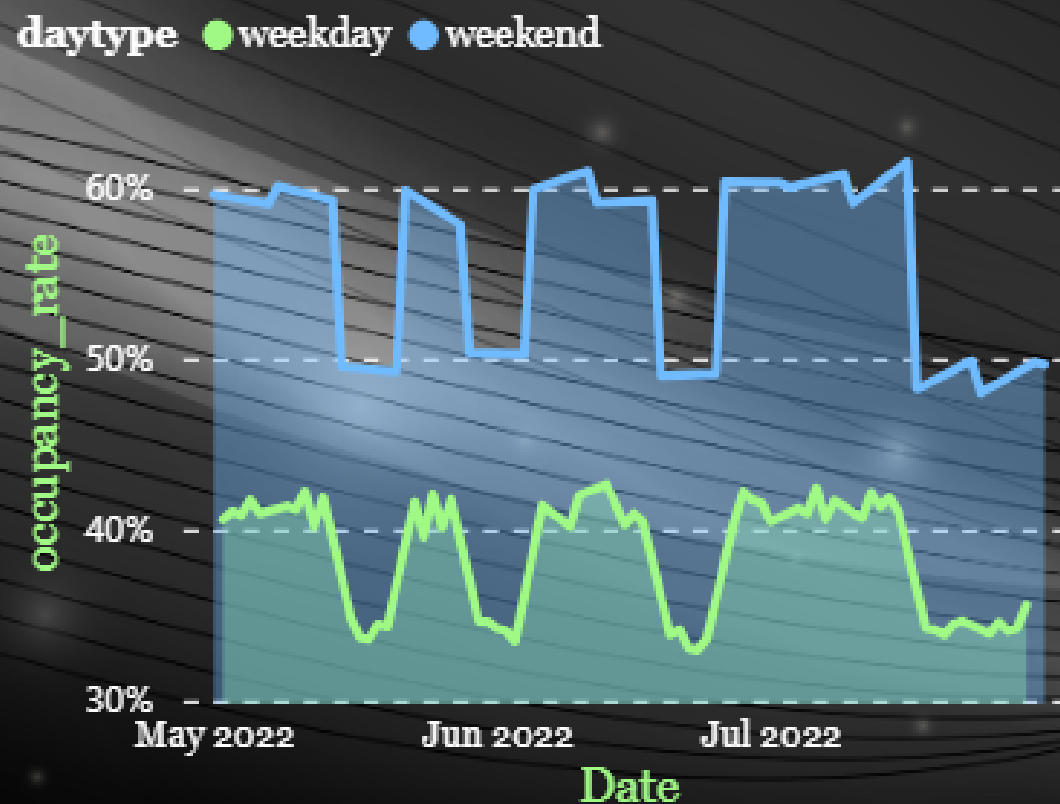
All

Day type

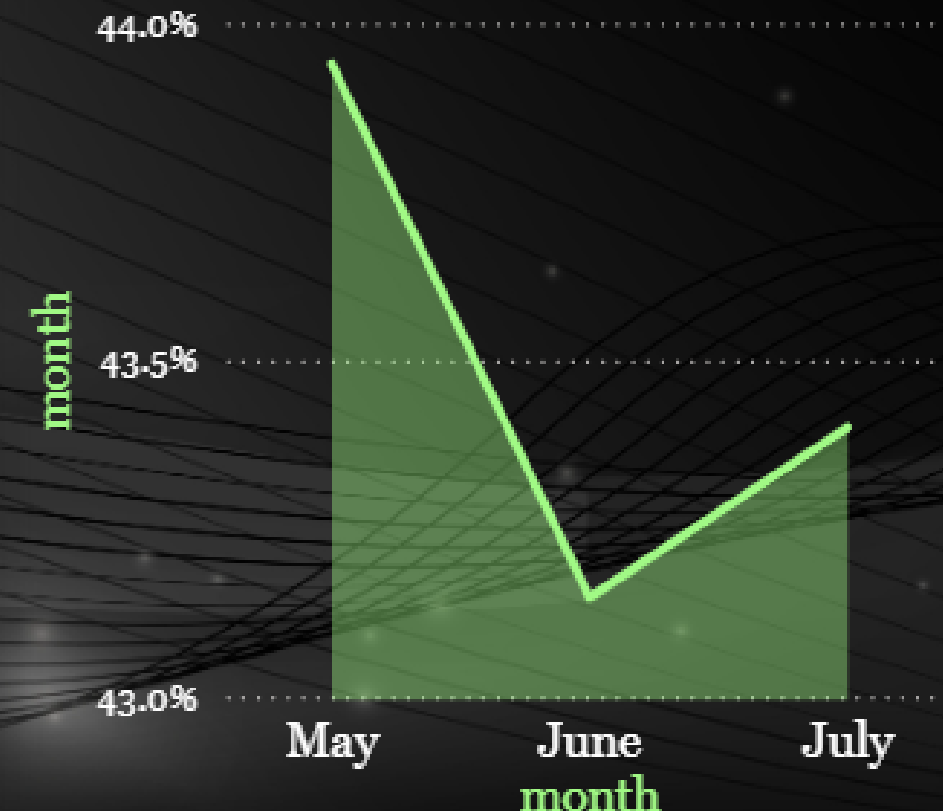
weekday

weekend

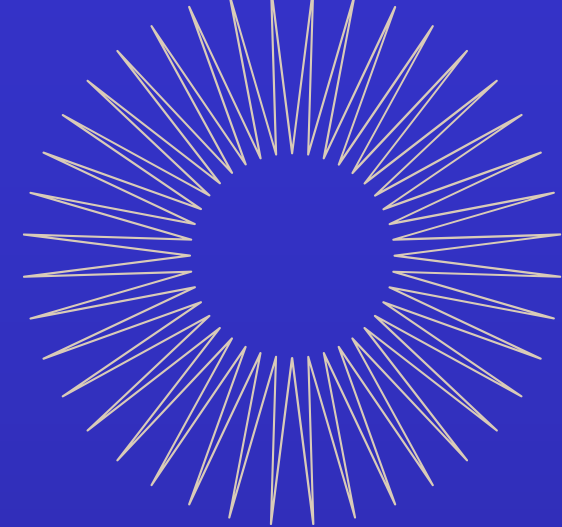
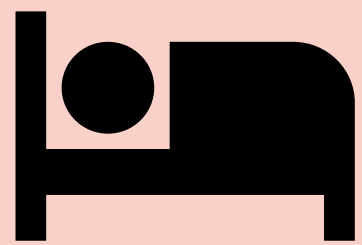
occupancy_rate by Date and daytype



occupancy_rate by Month



Key Insights: Occupancy & Capacity Analysis



Overall Performance

- OCCUPANCY RATE: 43.50% ACROSS 233K AVAILABLE ROOMS WITH REVPAR OF ₹7.35K
- GROWTH TRENDS: WOW GROWTH OF 13.14%, BUT MOM DECLINE OF -0.12% INDICATES SHORT-TERM MOMENTUM

Occupancy by Hotel & Location

- TOP PERFORMERS: AGRA LEADS AT 49.83%, FOLLOWED BY GOA (49.67%) AND DELHI (45.37%)
- UNDERPERFORMER: MUMBAI AT 41.44%, SUGGESTING PRICING OR DEMAND CHALLENGES IN THAT MARKET

Seasonal Fluctuation

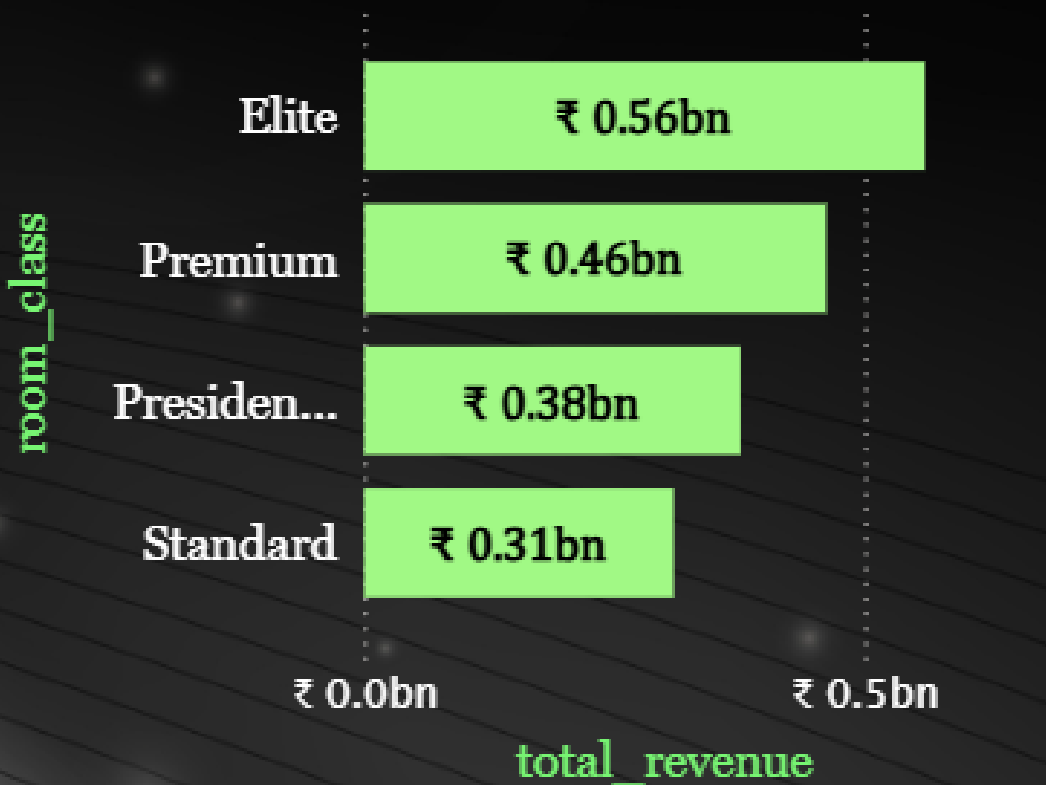
- WEEKDAY VS WEEKEND: WEEKENDS SHOW CONSISTENTLY LOWER OCCUPANCY (~50-55%) VERSUS WEEKDAYS (~40-45%), INDICATING LEISURE-DRIVEN DEMAND PATTERNS
- MONTHLY TRENDS: SHARP OCCUPANCY SPIKE IN EARLY MAY (44%), DECLINING TO 43% IN JUNE, THEN RECOVERING SLIGHTLY IN JULY—SEASONAL DIPS DURING MID-SUMMER MONTHS

Occupancy-Revenue Correlation

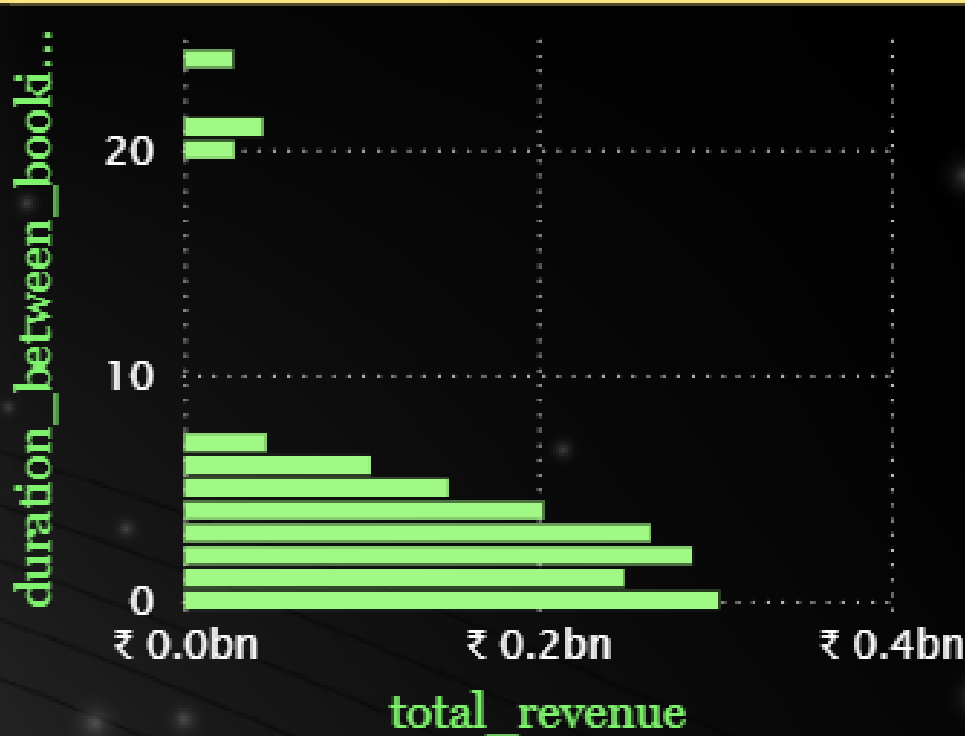
- STRONG POSITIVE LINK: SCATTER PLOT REVEALS HIGHER OCCUPANCY DIRECTLY DRIVES INCREASED REVPAR AND TOTAL REVENUE—PROPERTIES WITH 50%+ OCCUPANCY GENERATE ₹0.25-0.3BN REVENUE VERSUS ₹0.1-0.15BN AT 30-40% OCCUPANCY

Room Category Performance & Booking Insights

Revenue contribution by room_class



Effect of Booking Lead Time on Revenue and Occupancy



Date slicer

01-05-2022

31-07-2022

Hotel names

All

Room category

All

Day type

weekday

weekend

2

avg_IOS

3.71

Booking_lead_time

135K

total_booking

BOOKING

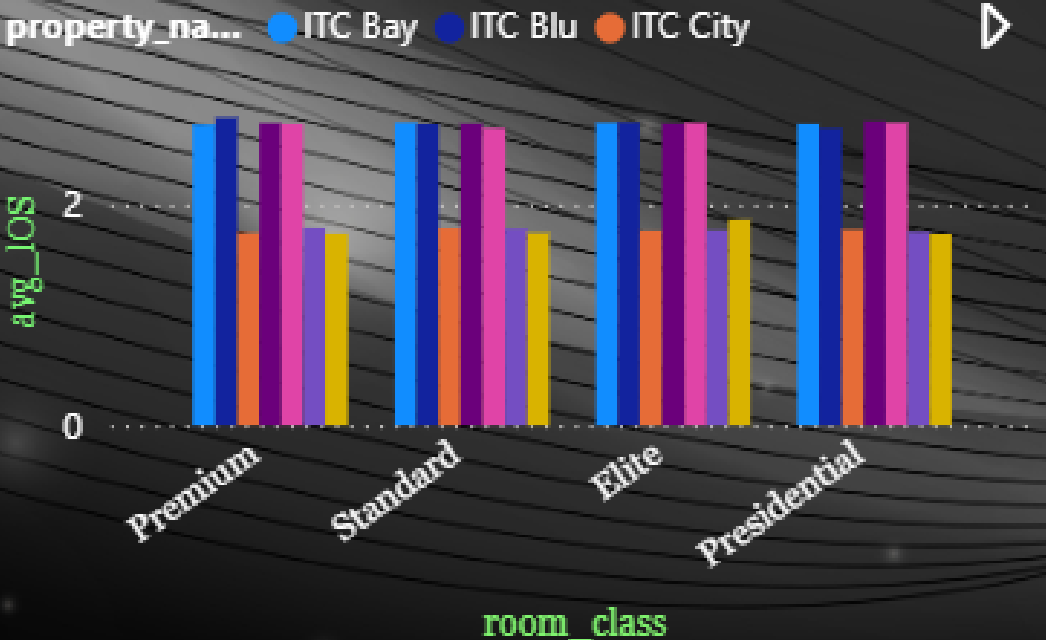
1388.75%

wow_growth

50.44%

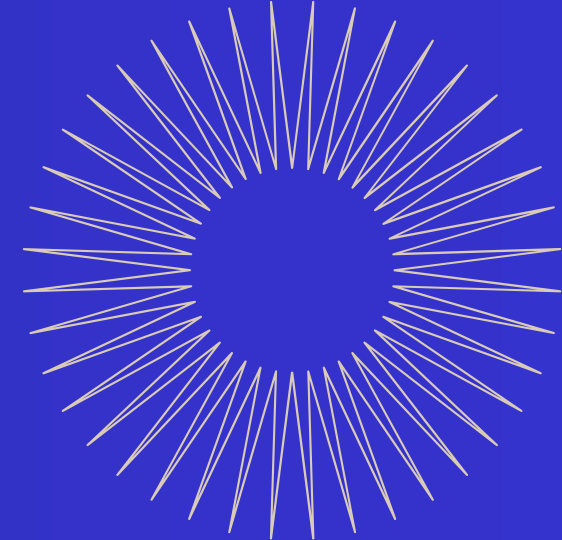
mom_growth

avg_IOS by room_class and property_name



Date	running_total_revenue
01 May 2022	₹2,53,57,111
02 May 2022	₹4,27,65,917
03 May 2022	₹6,03,00,105
04 May 2022	₹7,76,32,159
05 May 2022	₹9,55,33,195
06 May 2022	₹11,29,60,081
07 May 2022	₹13,81,82,064
08 May 2022	₹16,38,79,383
09 May 2022	₹18,14,17,364
10 May 2022	₹19,91,14,360
Total	₹1,70,87,71,229

Room Category Insights



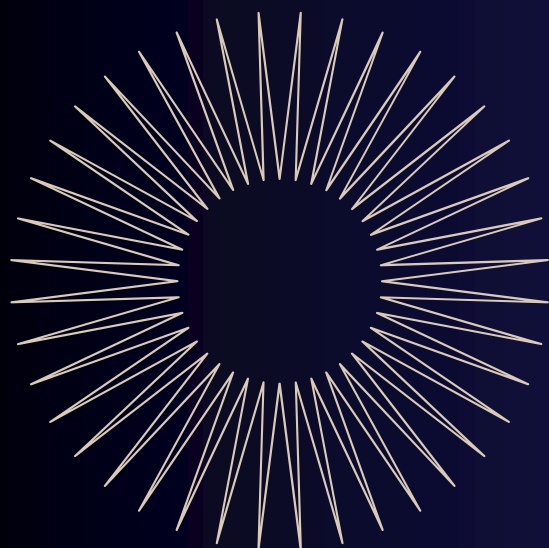
Quantitative Insights – Room Category & Booking Performance

- REVENUE CONTRIBUTION BY CATEGORY: ELITE: ₹0.56B (32.9%) – HIGHEST CONTRIBUTOR
- PREMIUM: ₹0.46B (27.1%)
- PRESIDENTIAL: ₹0.38B (22.4%)
- STANDARD: ₹0.31B (18.2%) – LOWEST
- TOP 2 CATEGORIES CONTRIBUTE ~60% OF TOTAL REVENUE

- BOOKING PATTERN IMPACT: AVG LEAD TIME: 3.71 DAYS → 70–80% BOOKINGS WITHIN 0–5 DAYS
- PEAK REVENUE (~₹0.25B–₹0.30B) COMES FROM SHORT LEAD-TIME BOOKINGS, SHOWING LAST-MINUTE DEMAND DOMINANCE

- AVERAGE LENGTH OF STAY (ALOS): OVERALL ALOS ≈ 2 DAYS
- PREMIUM/ELITE ROOMS SLIGHTLY HIGHER (~2.2–2.5 DAYS) VS STANDARD (~1.8–2 DAYS)
- HIGHER CATEGORIES = LONGER STAYS + HIGHER REVENUE

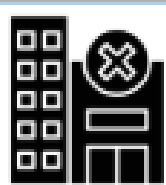
- REVENUE TREND & GROWTH: TOTAL REVENUE: ₹1.70B+
- RUNNING TOTAL: STEADILY INCREASING → NO DECLINE PHASES
- WOW GROWTH: 1388.75% (VERY HIGH SPIKE)
- MOM GROWTH: 50.44% → STRONG UPWARD MOMENTUM



Cancellations & Lost Revenue Analysis

Elite

Top Cancelling Room



\$5.72

avg_loss/cancellation

5

Peak Cancellation
Month

24.83%

Cancellation Rate %

33K

cancelled_bookings



299M

lost_revenue



6.93%

wow_cancellation_gr...

Date	cancelled_bookings	cumulative_cancellation
01 May 2022	498	498
02 May 2022	362	860
03 May 2022	347	1207
04 May 2022	354	1561
05 May 2022	358	1919
06 May 2022	363	2282
07 May 2022	497	2779
08 May 2022	493	3272
09 May 2022	345	3617
10 May 2022	354	3971
11 May 2022	337	4308
12 May 2022	366	4674
13 May 2022	328	5002
14 May 2022	488	5490
15 May 2022	410	5900
16 May 2022	269	6169
17 May 2022	281	6450
Total	33420	33420

Date slicer

01-05-2022

31-07-2022

Room class

All

Cancellation Rate % by
property_name

property_name

ITC P... 25.18%

ITC G... 25.08%

ITC City 24.92%

ITC Bay 24.84%

ITC S... 24.79%

ITC Blu 24.65%

ITC E... 24.37%

0%

20%

Cancellation Rate %

Hotel names

ITC Bay

ITC City

ITC
Grands

ITC Blu

ITC
Exotica

ITC
Palace

Cancellation parameter

city

room_category

property_name

room_class

Booking platform

All



Cancellation Insights



Cancellations & Lost Revenue – Quantitative Insights

- OVERALL CANCELLATION IMPACT: CANCELLATION RATE: 24.83% → ~1 IN 4 BOOKINGS GETS CANCELLED
- TOTAL CANCELLATIONS: 33K BOOKINGS
- LOST REVENUE: ₹299M (~₹0.3B) → SIGNIFICANT REVENUE LEAKAGE
- CANCELLATION BY PROPERTY: HIGHEST: ~25.18% (ITC PALACE)
- LOWEST: ~24.37% (ITC EXOTICA)
- VERY NARROW RANGE (~24.3%–25.2%) → CANCELLATIONS ARE CONSISTENT ACROSS ALL HOTELS (SYSTEM-WIDE ISSUE)
- TREND OVER TIME: WOW CANCELLATION GROWTH: 6.93% → CANCELLATIONS ARE INCREASING GRADUALLY
- DAILY CANCELLATIONS RANGE ~300–500 BOOKINGS, SHOWING STABLE BUT HIGH VOLUME
- CUMULATIVE CANCELLATIONS STEADILY RISE TO 33K, INDICATING CONTINUOUS LOSS BUILDUP
- REVENUE LOSS PER CANCELLATION: AVG LOSS PER BOOKING: ~\$5.72 (~₹475 APPROX)
- EVEN SMALL PER-BOOKING LOSS LEADS TO LARGE TOTAL IMPACT (₹299M)
- PATTERNS IDENTIFIED: ELITE ROOMS ARE MOST CANCELLED, DESPITE BEING HIGH-REVENUE → MAJOR PROFIT LEAKAGE
- NO STRONG VARIATION BY PROPERTY → ISSUE LIKELY DUE TO BOOKING BEHAVIOR (LAST-MINUTE CANCELLATIONS / FLEXIBLE POLICIES)
- PEAK CANCELLATION PERIOD IDENTIFIED AROUND MAY (HIGH BOOKING ACTIVITY PERIOD)



Final Business Insights – ITC Hotels

💰 **Revenue Performance**

- TOTAL REVENUE SHOWS STRONG GROWTH TREND, WITH PEAK PERFORMANCE IN MAY–JULY
- ELITE & PREMIUM ROOMS ARE THE HIGHEST REVENUE CONTRIBUTORS
- HIGH ADR (₹14.92K) AND REVPAR (₹7.35K) INDICATE EFFECTIVE PRICING STRATEGY

🏠 **Occupancy & Demand Patterns**

- AVERAGE OCCUPANCY ~43.5%
- WEEKENDS PERFORM WELL INSTEAD OF WEEKDAYS IN OCCUPANCY
- CITIES LIKE AGRA & GOA SHOW HIGHER DEMAND VS METRO CITIES

🛏️ **Customer Behavior & Booking Trends**

- AVERAGE LENGTH OF STAY -2 DAYS
- BOOKING LEAD TIME -3.7 DAYS → MAJORITY ARE LAST-MINUTE BOOKINGS
- LUXURY CATEGORIES SHOW LONGER STAYS & HIGHER SPENDING BEHAVIOR

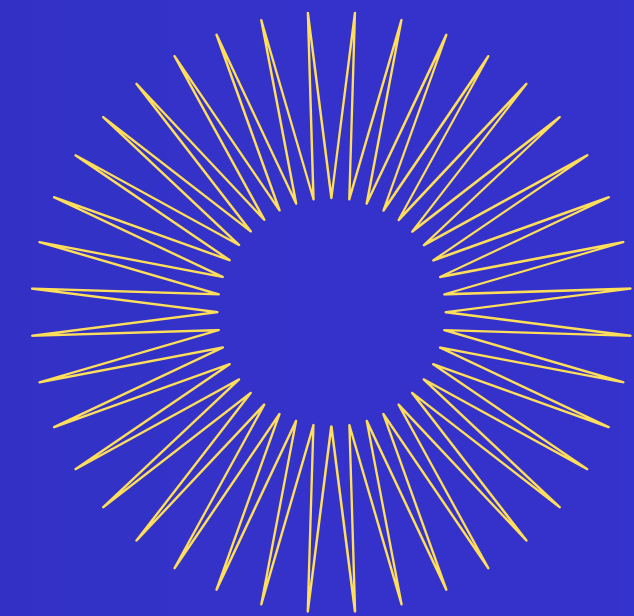
❌ **Cancellations & Revenue Leakage**

- CANCELLATION RATE IS HIGH (24.83%)
- SIGNIFICANT LOST REVENUE (₹299M) DUE TO CANCELLATIONS
- STANDARD ROOMS HAVE HIGHER CANCELLATIONS, WHILE LUXURY ROOMS SHOW LOWER CANCELLATION

🌟 **Luxury Ambiance Impact**

- LUXURY ROOMS CONTRIBUTE HIGHER REVENUE DESPITE MODERATE OCCUPANCY
- CUSTOMERS BOOKING PREMIUM ROOMS SHOW:
- LOWER CANCELLATION RATES
- HIGHER RETENTION & LONGER STAYS

INDICATES LUXURY EXPERIENCE & HIGH VALUE CUSTOMER SEGMENT



Final Conclusion– ITC Hotels

◆ Business Performance Summary

- ITC HOTELS DEMONSTRATES STRONG REVENUE PERFORMANCE (~₹1.7BN+) DRIVEN PRIMARILY BY PREMIUM & ELITE ROOM CATEGORIES
- STABLE REVPAR AND ADR INDICATE EFFECTIVE PRICING STRATEGIES
- OCCUPANCY (~43–50%) SHOWS BALANCED DEMAND, WITH HIGHER UTILIZATION DURING WEEKENDS AND PEAK SEASONS

◆ Customer Behavior Insights

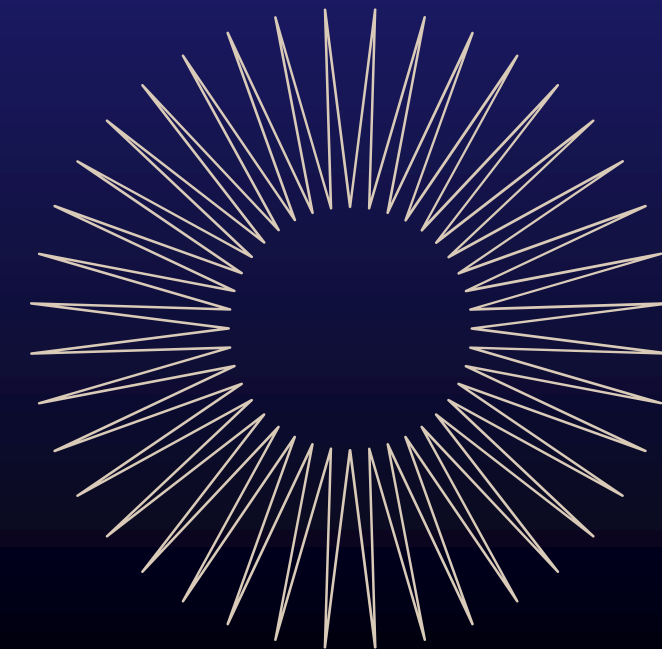
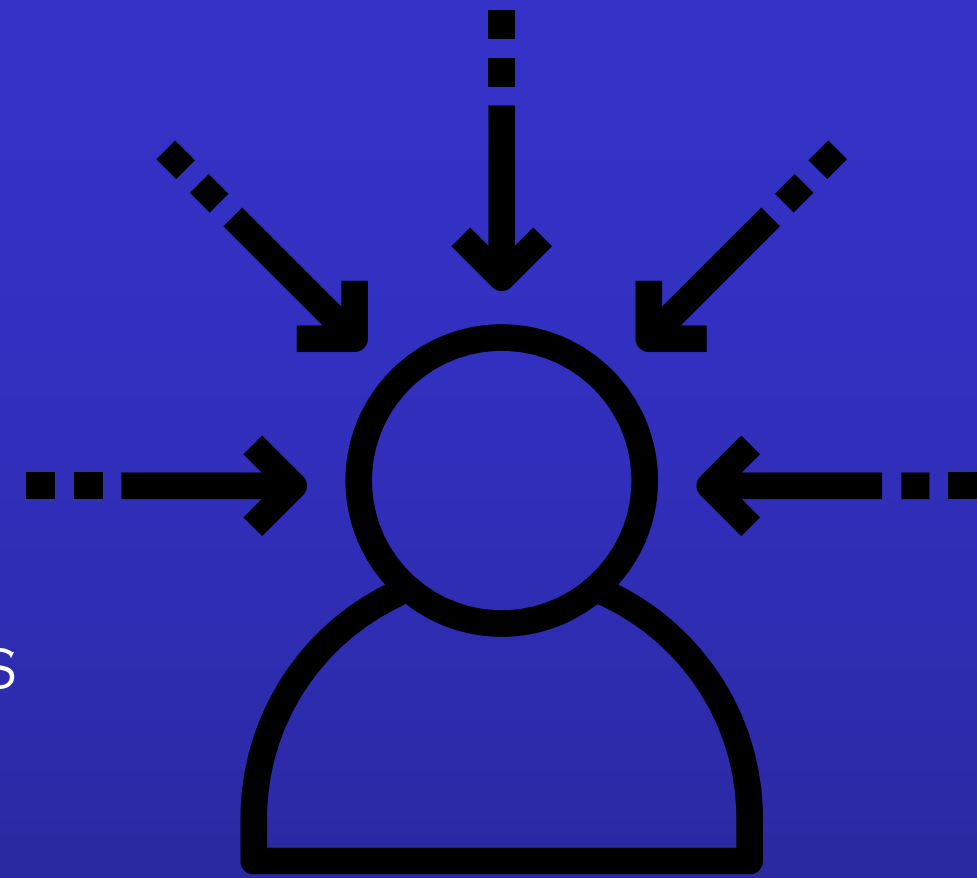
- GUESTS BOOKING LUXURY ROOMS TEND TO STAY LONGER (HIGHER ALOS)
- EARLY BOOKINGS (HIGHER LEAD TIME) ARE LINKED TO HIGHER-VALUE CUSTOMERS
- LOWER-TIER ROOMS SHOW HIGHER CANCELLATION RATES, IMPACTING REVENUE STABILITY

◆ Cancellation & Revenue Impact

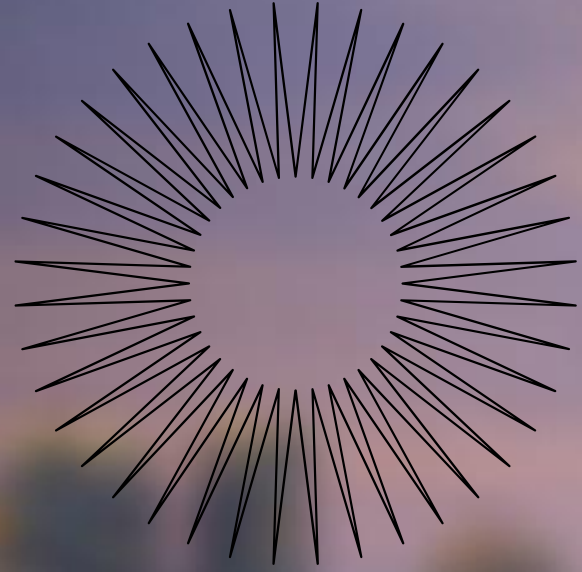
- SIGNIFICANT LOST REVENUE (~₹299M) DUE TO CANCELLATIONS
- LUXURY SEGMENT SHOWS LOWER CANCELLATION TENDENCY

◆ Guest Experience & Luxury Ambiance Insight

- ITC HOTELS SUCCESSFULLY DELIVERS A PREMIUM LUXURY EXPERIENCE THAT:
 - DRIVES HIGHER WILLINGNESS TO PAY (HIGHER ADR)
 - ENCOURAGES LONGER STAYS AND REPEAT BEHAVIOR
 - ATTRACTS HIGH-VALUE EXPERIENCE-FOCUSED CUSTOMERS



Thanks for visiting...



ITC HOTELS

RESPONSIBLE LUXURY



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